

Measuring the prompts, not trusting them

Three stories, seven regeneration rounds, four prompt revisions, two reverts, and one new output format.

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The one-line result. The complaint that started this — video frames being "really good variations of the same image" — was not where it appeared to be. The script's declared visual subjects were already near-perfectly distinct (0.028 pairwise overlap); the art-director step turned them into near-repetitive scenes (0.327). A ten-fold collapse, one step downstream. Fixing it cut the share of frames containing a desk, laptop or screen from **0.413 to 0.315**, confirmed at **0.355 on real end-to-end renders**.

Method

Every metric is computed without a model and without a human, so a prompt change is judged by whether a number moved rather than by whether the output felt better. The three test stories were chosen to be deliberately unlike — a consumer hardware launch, a systems-internals piece about `malloc()`, and a web-history story with no hardware in it — so a fix that only works on one kind of story shows up as failing to generalise.

Calibration came first, and changed the conclusion

Raw pairwise overlap cannot be read on its own. Scenes drawn from three *different stories* — definitionally unrelated — still score **0.161**, because cinematic prose shares a great deal of unavoidable vocabulary. That is the floor, not zero. Identical scenes score 1.000. Without measuring the floor the headline number would have looked like a failure; with it, the picture is that overlap was the wrong instrument and a second metric was needed.

What changed, and what each change did

1 · The art director (image.scene v3 → v5)

Two causes, both instructions I had written myself in an earlier session. `image.scene` said "*keep the same world, cast and place where the story has not moved on*" — added for cross-shot continuity, and equally an instruction to repeat the subject. And `prior_scenes` passed the full text of previous prompts, so the model simply continued the prose.

- **Split continuity in two.** Hold era, palette, scale and indoors/outdoors; change what is in front of the camera.
- **Pass prior subjects, not prior scenes** — a bare noun list, framed as things to avoid.
- **A binding visual register per beat**, rotating through human scale, material, architectural, mechanism, landscape, trace. Camera angle was never the problem: shot coverage already measured 0.64 while subjects repeated, so a take could show eight angles of one desk.
- **Demoted the spoken line** from the top of the template to context at the bottom. It was anchoring every frame on hardware.

METRIC	V3 BASELINE	V4	V5	RENDERED
Tech-tableau rate	0.413	0.319	0.315	0.355
Scene vocabulary	0.460	0.495	0.502	0.437
Excess overlap (above 0.161 floor)	0.157	0.156	0.150	0.102
Distinct head nouns	0.891	0.958	0.917	—

Tableau rate is the share of frames containing a desk, laptop, screen, monitor, keyboard, phone or office — the number that tracks what a viewer actually notices. Excess overlap barely moved, and that is reported rather than buried: it is dominated by prose style rather than subject, so it was the wrong instrument for this failure. Both are kept, because a future change that games one will show up in the other.

2 · The script (script.write v1 → v5, and a guard)

Baseline script metrics were mostly healthy. The visible defects were a rich article rendered as **six** sections against a requested 10–16, and section lengths clustering.

ATTEMPT	CHANGE	SECTIONS	LENGTH VARIATION
v1	baseline	16 / 6 / 9	0.224
v3	numeric rhythm quota + firmer floor	14 / 15 / 8	0.139
v4	function-driven rhythm, softer floor	10 / 12 / 3	0.146
v5	v3 floor + plain rhythm line	14 / 14 / 6	0.165
guard	retry under 8 sections	14 / 12 / 12	0.206

Two changes were reverted, and they are the more useful findings. Asking for "at least two sections under ten words and two over twenty-five" produced *fewer* of both — short sections went from 2/0/0 to 0/0/0. A numeric target made the model aim for the middle. Softening the section floor to a function-driven framing collapsed one script to three sections.

The real finding is that section count is not a prompt problem at all. The same story and prompt returned **9, 8, 3 and 6** sections across four samples. It is unstable run to run, and no wording made it reliable. The floor is now enforced in code: under eight sections from more than 3,000 characters of source, retry once and keep the fuller script.

The rule that emerged

Across three rounds the same division held, and it is now the working rule for this system:

Structural properties need a deterministic guard. Speaker run length, teaser subject, section count — wording is unreliable for all three, and a cheap post-generation check is not. All three are now guarded in code.

Content properties can only be fixed in the prompt. Which subject to draw, what register to work in, what to avoid repeating — no validator can supply these, and the prompt moved them measurably.

New capability: narrated digests

An edition can now be voiced as one continuous listen, built from the same composed sections the Kindle document is typeset from — deliberately not a separate audio script, which would drift from the written edition within a week. Voices alternate per item, so it reads as two presenters trading pieces rather than one monologue.

The first attempt voiced 3 of 5 sections. The cause is a hard backend limit worth recording: **Magpie returns audio over gRPC with a 4 MB response cap**, roughly 95 seconds or 350 words at 22050 Hz. Script lines never reach it; digest sections are 350–500 words and reach it every time. Length was not obvious as the cause because 1,584 characters worked fine in isolation. Narration is now chunked on sentence boundaries — a mid-sentence split is audible, because the voice drops its intonation contour at the break.

SAME EDITION	BEFORE	AFTER
Sections voiced	3 of 5	12 of 12
Duration	2:09	10:45

Round-by-round record

ROUND	SWITCH	SCRIPT VOCAB	LENGTH VAR	INTENT OVERLAP
round1-baseline	0.456	0.748	0.224	0.028
round3-script	0.511	0.727	0.139	0.029
round3b-script	0.572	0.717	0.146	0.049
round3c-script	0.585	0.716	0.165	0.048
round3d-guard	0.457	0.727	0.206	0.021

SCENE ROUND	SUBJECT OVERLAP	WORST PAIR	VOCABULARY	SHOT COVERAGE
round1-scenes (v4)	0.317	0.392	0.495	0.727
round2-scenes (v5)	0.311	0.389	0.502	0.863

What I would do next

- **The register rotation deserves a second pass.** It moved tableau rate 24% as a fixed six-item cycle. Choosing the register from the beat's own `visual_intent` — mechanism for a how-it-works beat, landscape for a consequence beat — should do better than rotating blind.
- **Video digests** are the obvious counterpart to the narrated edition: the same composed sections, one card per story, over the existing image pipeline. The audio side proved the shape works.
- **Length variation is still unsolved.** Neither prompt approach produced short sections, and unlike section count it has no obvious deterministic fix — a validator can detect uniformity but cannot write the one-line objection that breaks it.
- **An ASR loop back into scoring.** The pipeline already transcribes its own audio and scored 0.948 on the last run; that signal is computed and then discarded.